Machine Learning Libraries and Frameworks

Choose among no-code AutoML, managed ML services, SQL-based modeling, and open-source Python frameworks according to skill, control, scale, and operational needs.

Original study notes based on Gwendolyn Stripling and Michael Abel, First edition, 2023. Book pp. 49–70. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

Business need → AutoML / SQL ML / Python → Compare control + cost → Validate on held-out data

Detailed Notes

No-code AutoML and how it works

Source: book pp. 49–61 / PDF p. 65

Typical AutoML flow: upload a labeled dataset, identify the target, inspect schema/statistics, choose a task objective, define a split/budget, train candidate pipelines, review metrics/attributions, and deploy if it meets requirements.

Automation can test model families, preprocessing, and hyperparameters. It cannot determine whether the target is appropriate, the split is leakage-free, or a score maps to business value. Inspect class imbalance, label quality, feature availability, and the evaluation set. Understand service quotas, region, pricing, and data handling before upload.

Machine learning as a service

Source: book pp. 62–65 / PDF p. 78

Managed cloud services provide datasets, training jobs, model registries, endpoints, and monitoring. They reduce infrastructure setup but introduce provider-specific IAM, quotas, formats, costs, and operational dependencies.

Choose managed services for time-to-value when controls and regional/privacy requirements fit. Keep records of project, dataset, model, endpoint, and deployment versions. Undeploy endpoints when idle if charges continue.

Low-code, SQL, and open-source frameworks

Source: book pp. 66–70 / PDF p. 82

SQL ML lets analysts use familiar queries for data preparation and model training close to warehouse data. It reduces data movement but depends on supported model types and SQL-specific semantics. BigQuery ML is one example.

Scikit-learn offers composable preprocessing, estimators, pipelines, and metrics; Keras/TensorFlow supports neural networks; AutoKeras, auto-sklearn, and Auto-PyTorch add automated search at different layers. More abstraction saves code but can reduce transparency/control. Pick the least complex option that can be audited, evaluated, and maintained.

Revision Check

1. Which steps does AutoML automate, and which remain your responsibility?
2. Would SQL ML reduce unnecessary data movement here?
3. What controls would require custom code?
4. Who owns deployment cost and endpoint cleanup?

No-code, low-code, and custom code are points on a control spectrum. Choose based on the problem and constraints, not prestige.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.